

Paradoxical dominant negative activity of an immunodeficiency-associated activating *PIK3R1* variant

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Abstract

PIK3R1 encodes three regulatory subunits of class IA phosphoinositide 3-kinase (PI3K), each associating with any of three catalytic subunits, namely p110 α , p110 β or p110 δ . Constitutional *PIK3R1* mutations cause diseases with a genotype-phenotype relationship not yet fully explained: heterozygous loss-of-function mutations cause SHORT syndrome, featuring insulin resistance and short stature attributed to reduced p110 α function, while heterozygous activating mutations cause immunodeficiency, attributed to p110 δ activation and known as APDS2. Surprisingly, APDS2 patients do not show features of p110 α hyperactivation, but do commonly have short stature or SHORT syndrome, suggesting p110 α hypofunction. We sought to investigate this. In dermal fibroblasts from an APDS2 patient, we found no increased PI3K signalling, with p110 δ expression markedly reduced. In preadipocytes, the APDS2 variant was potently dominant negative, associating with Irs1 and Irs2 but failing to heterodimerise with p110 α . This attenuation of p110 α signalling by a p110 δ -activating *PIK3R1* variant potentially explains co-incidence of gain-of-function and loss-of-function *PIK3R1* phenotypes.

eLife assessment

The authors identify new mechanisms that link a *PIK3R1* mutant to cellular signaling and division in Activated PI3 Kinase Delta Syndrom 1 and 2 (APDS1/2). The conclusion that this mutant serves as a dominant negative form of the protein, impacting PI3K complex assembly and IRS/AKT signaling, is **important**, and the evidence from constitutive and inducible systems in cultured cells is **convincing**. Nevertheless, there are several limitations relating to differences between cell lines and expression systems, as well as more global characterization of the protein interaction landscape, which would further enhance the work.

Introduction

PIK3R1 gene products (p85 α , p55 α and p50 α) are essential for class IA phosphoinositide 3- kinase (PI3K) signalling. Each of the three protein products can bind any of three catalytic subunits, p110 α , β and δ , encoded by *PIK3CA*, *PIK3CB*, and *PIK3CD* respectively. This stabilises and inhibits the catalytic subunits in the basal state and confers responsiveness to upstream stimuli including receptor tyrosine kinase activation, enabled by two phosphotyrosine-binding SH2 domains shared by all *PIK3R1* products (Fruman *et al*, 2017 [↗](#)).

PI3K signalling mediates responses to a myriad of cues, including growth factors, hormones such as insulin, and processed antigens, and so recent discovery that genetic perturbation of *PIK3R1* in humans disrupts growth, insulin action and immunity is no surprise. Beyond this headline observation lies considerable complexity, however, and important questions about genotype-phenotype correlations in *PIK3R1*-related disorders are unresolved.

Some *PIK3R1* mutations reduce basal inhibition of catalytic subunits, usually due to disruption of the inhibitory inter-SH2 domain, and are found in cancers (Philp *et al*, 2001 [↗](#)) and vascular malformations with overgrowth (Cottrell *et al*, 2021 [↗](#)). In both diseases, hyperactivated PI3K α , composed of heterodimers of *PIK3R1* products and p110 α , drives pathological growth. Distinct inter-SH2 domain *PIK3R1* mutations, mostly causing skipping of exon 11 and deletion of residues 434-475, hyperactivate PI3K δ in immune cells, causing immunodeficiency (Deau *et al*, 2014 [↗](#); Lucas *et al*, 2014b [↗](#)). This phenocopies the immunodeficiency caused by genetic activation of p110 δ itself, called Activated PI3K Delta Syndrome 1 (APDS1) (Angulo *et al*, 2013 [↗](#); Lucas *et al*, 2014a [↗](#)), and is thus named APDS2.

Despite ubiquitous *PIK3R1* expression, features attributable to PI3K α activation have been described in only a single case of APDS2 to date (Wentink *et al*, 2017 [↗](#)). In that case, the causal variant was a constitutional activating mutation associated with cancer rather than one of the common APDS2 variants. Biochemical studies have suggested that apparently selective p110 δ activation occurs because APDS2 mutations derepress p110 δ , the predominant immune system catalytic subunit, more than p110 α , due to differences in the inhibitory contacts between *PIK3R1* products and different catalytic subunits (Dornan *et al*, 2017 [↗](#)).

More surprisingly, phenotypic overlap is reported between APDS2 and SHORT syndrome. SHORT syndrome is caused by loss of PI3K α function due to disruption of the phosphotyrosine-binding C-terminal SH2 domain (Chudasama *et al*, 2013 [↗](#); Dymment *et al*, 2013 [↗](#); Thauvin-Robinet *et al*, 2013 [↗](#)). It features reduced linear growth, insulin resistance, and dysmorphic features (Avila *et al*, 2016 [↗](#)). In recent years, both case reports (Bravo Garcia- Morato *et al*, 2017; Petrovski *et al*, 2016 [↗](#); Ramirez *et al*, 2020 [↗](#); Szczawinska-Poplonyk *et al*, 2022 [↗](#)) and larger series (Elkaim *et al*, 2016 [↗](#); Jamee *et al*, 2020 [↗](#); Maccari *et al*, 2023 [↗](#); Nguyen *et al*, 2023 [↗](#); Olbrich *et al*, 2016 [↗](#); Petrovski *et al*, 2016 [↗](#)) have established that many people with APDS2 have overt features of SHORT syndrome, while, more generally, linear growth impairment is common in APDS2, but not in APDS1. These clinical observations are bolstered by the impaired linear growth and increased *in utero* mortality reported in mice with knock in of the common causal APDS2 mutation in *Pik3r1* (Nguyen *et al*, 2023 [↗](#)). All people with SHORT-APDS2 overlap syndromes have well established activating *PIK3R1* mutations in the inter-SH2 domain implicated in APDS2, but none have characteristic SHORT syndrome mutations, which are usually in the C-terminal SH2 domain. Conversely, no patient with a classical SHORT syndrome mutation has been shown to have immunodeficiency. There is thus now convincing evidence of a syndrome with features of both gain and loss of PI3K function. The mechanistic basis of this is unexplained.

PI3K activity is determined not just by activity of individual PI3K holoenzymes, but also by subunit stoichiometry. Regulatory and catalytic subunits stabilise each other upon binding (Brachmann *et al*, 2005 [↗](#); Yu *et al*, 1998 [↗](#)), and only regulatory subunit monomers are stable enough to be detected in mammalian cells (Yu *et al*, 1998 [↗](#)). Molar excess of regulatory subunits over catalytic subunits gives rise to free regulatory subunits that compete with holoenzyme for phosphotyrosine binding (Ueki *et al*, 2002 [↗](#)). This has been invoked to explain increased insulin sensitivity on knockout or reduction of p85α alone (Mauvais-Jarvis *et al*, 2002 [↗](#)), of p55α and p50α (Chen *et al*, 2004 [↗](#)) or of all *Pik3r1* products (Fruman *et al*, 2000 [↗](#)). The “free-p85” model has been contested by some reports suggesting no excess p85 (Geering *et al*, 2007 [↗](#); Zhao *et al*, 2006 [↗](#)), however most observations, including some made recently using *in vivo* tagging and pulldown of PI3K components (Tsolakos *et al*, 2018 [↗](#)) suggest that monomeric p85α can be seen *in vivo* in some tissues.

We now address mechanisms underlying human mixed gain- and loss-of function phenotypes observed in association with PIK3R1 mutations using primary cells, cell models and *in vitro* enzyme assay and suggest that they are explained by competing effects of APDS2 PIK3R1 mutations on PI3K activity and stability.

Results

PI3K subunit expression and signalling in APDS2 fibroblasts

Disorders in the *PIK3CA*-related overgrowth spectrum (PROS) feature soft tissue overgrowth and are caused by mosaic activating mutations in *PIK3CA*, encoding the p110α catalytic subunit of PI3Kα (Madsen *et al*, 2018 [↗](#)). In this group, basal hyperactivation of PI3Kα signalling can be easily discerned in dermal fibroblasts from affected areas of the body (Lindhurst *et al*, 2012 [↗](#)). *PIK3R1*, like *PIK3CA*, is ubiquitously expressed, yet overgrowth is not reported in APDS2 caused by constitutional activating mutations in *PIK3R1*. Dermal fibroblasts strongly express *PIK3R1*, but no studies to date have evaluated whether PI3K activity is increased in APDS2.

We began by assessing dermal fibroblasts cultured from a previously described woman with APDS2 due to the common causal PIK3R1 mutation in the splice donor site that causes skipping of exon 11 and an in-frame deletion in the inter-SH2 domain shared by all PIK3R1 isoforms (Patient A.1 in (Lucas *et al*, 2014b [↗](#))). These cells were compared to cells from healthy control volunteers, or from people with PROS. Confirmation of expression of all pathogenic mutations was undertaken by cDNA sequencing prior to further study (Fig 1 supplement 1A [↗](#)). We found that truncated p85α was expressed, but at a much lower level than full length wild-type p85α (Fig 1A [↗](#), Fig 1 supplement 1B,C [↗](#)). Despite this, no increase in basal or insulin-stimulated AKT phosphorylation was seen in APDS2 cells compared to cells from wild-type volunteers or from people with PROS and activating *PIK3CA* mutations H1047L or H1047R (Fig 1A-C [↗](#), Fig 1 supplement 2A,B [↗](#)). Indeed, insulin-induced phosphorylation of AKT was lower in fibroblasts from this APDS2 patient compared to controls.

Previous studies have suggested that the truncated, APDS2-causal p85α variant exerts a much greater activating effect on p110δ, which has a more restricted tissue expression, including immune cells, than p110α, which is ubiquitous (Dornan *et al*, 2017 [↗](#)). We found that both p110α and p110δ were easily detectable in control dermal fibroblasts, however p110δ was almost absent in APDS2 fibroblasts compared to controls, with lower levels also seen in PROS cells (Fig 1A,D [↗](#), Fig 1 supplement 2D [↗](#)). p110α was unchanged in APDS2 cells but modestly increased in PROS cells (Fig 1A,E [↗](#), Fig 1 supplement 2C [↗](#)). Collectively these findings suggest that any ability of the APDS2 PIK3R1 variant in skin cells to activate PI3K may be overcome by reduced protein levels of p110δ, likely through reduced binding and/or reduced stabilisation of p110δ by the mutant regulatory subunit.

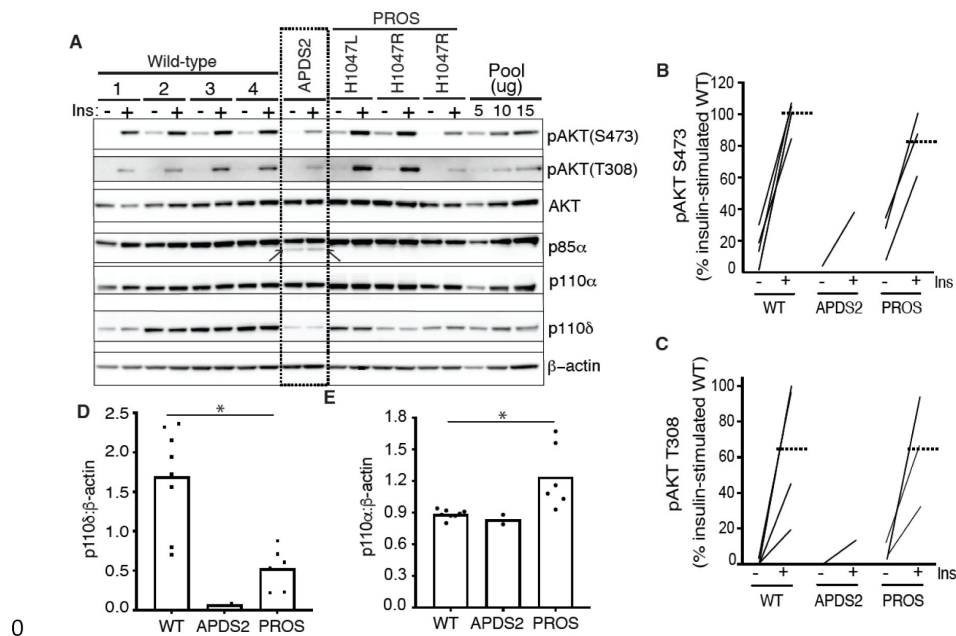


Figure 1.

PI3K subunit expression and signalling in primary dermal fibroblasts.

Immunoblotting of AKT, AKT phosphorylated at threonine 308 (T308) or serine 473 (S473), p85α, p110δ and p110α and are shown with and without stimulation by 100 nM insulin (Ins) for 10 minutes. β-actin is shown as a loading control, with different amounts of pooled lysate (Pool) used to demonstrate signal intensity in the linear range. Results from 4 healthy controls (WT; 1-4), 1 patient with Activating p110 Delta Syndrome 2 (APDS2) due to the p85α Δexon11 variant, and 3 patients with PIK3CA-Related Overgrowth Spectrum (PROS) caused by the activating *PIK3CA* mutations indicated. (A) Immunoblots, with the truncated p85α Δexon11 variant arrowed; (B)-(E) quantification of immunoblot bands from 3 independent experiments shown for phosphoAKT-S473, phosphoAKT-T308, p110δ and p110α respectively. Paired datapoints +/- insulin are shown in (B) and (C), and dotted lines mark means. Asterisks indicate a significant difference. More detailed statistical analysis including 95% confidence intervals for the paired mean differences for these comparisons are shown in Figure EV2.

Overexpressed PIK3R1 ΔExon11 is potently dominant negative in 3T3-L1 preadipocytes

We next turned to a well-established cellular system allowing conditional overexpression of *PK3R1* alleles of interest. We have previously shown that overexpression of two SHORT syndrome PIK3R1 variants – R649W and Y657X – impair insulin signalling and adipocyte differentiation of murine 3T3-L1 preadipocytes, consistent with impaired PI3Kα function and with the lipodystrophy and insulin resistance seen in SHORT syndrome (Huang-Doran *et al.*, 2016). To assess the effect of the APDS2 ΔEx11 in the same, non-immunological, cell context, we used lentiviral vectors, as previously described (Huang-Doran *et al.*, 2016), to generate clonal 3T3-L1 cell lines allowing conditional, tuneable overexpression of PIK3R1 variants in response to doxycycline. For signalling studies, we also generated cells conditionally overexpressing the PROS- and cancer-associated PIK3CA H1047R mutation as a positive control for increased PI3Kα signalling, while for differentiation studies we used previously reported lines conditionally expressing PIK3R1 R649W or Y657X (Huang-Doran *et al.*, 2016).

In undifferentiated cells, we first confirmed doxycycline-dependent overexpression of p85α or p110α transgenes (Fig 2A), before assessing basal and insulin-stimulated Akt phosphorylation. As expected, overexpression of oncogenic H1047R p110α strongly increased basal Akt phosphorylation (Fig 2A-C, Fig 2 supplement 1A,C) with no additional increase on insulin stimulation (Fig 2A-C, Fig 2 supplement 1B,D). Surprisingly, however, not only did overexpression of the APDS2 ΔEx11 p85α not increase basal PI3Kα signalling, but it potently inhibited insulin-induced Akt phosphorylation, consistent with a strong dominant negative action on pathway activation (Fig 2A-C, Fig 2 supplement 1). Overexpressing wild-type p85α mildly reduced insulin-stimulated Akt phosphorylation (Fig 2A-C, Fig 2 supplement 1B,C), although WT p85α was not expressed to as high a level as ΔEx11 p85α. In keeping with impaired PI3Kα activity, overexpression of the ΔEx11 variant also severely impaired adipocyte differentiation, assessed by triglyceride accumulation in response to a standard differentiation protocol (Fig 2D). A similar effect was seen on overexpressing SHORT syndrome variants R649W and Y657X (Fig 2D).

To assess whether the inhibitory effect of ΔEx11 p85α might be an artefact caused by strong overexpression, doxycycline titration was used to assess whether a low level of overexpression might unmask signalling hyperactivation. However, no such hyperactivation was seen, and instead, a graded diminution of insulin-induced AKT phosphorylation was observed from the lowest to highest levels of p85α ΔEx11 overexpression (Fig 2 supplement 2).

Effect of PIK3R1 mutations on Phosphoinositide 3-kinase activity in vitro

Given this evidence that APDS2-associated PIK3R1 ΔEx11 potently inhibits PI3Kα when overexpressed in 3T3-L1 preadipocytes, we next sought to investigate the biochemical basis of this observation. First, we assessed the effect of disease-causing PIK3R1 mutations on basal and phosphotyrosine-stimulated activity of purified PI3Kα, β and δ holoenzyme in a previously described reconstituted *in vitro* system (Burke *et al.*, 2012). As well as the APDS2 ΔEx11 mutation, we selected three SHORT syndrome-associated mutations to study. These were the most common causal mutation, R649W, which abolishes phosphotyrosine binding by the C-terminal SH2 (cSH2) domain (Chudasama *et al.*, 2013), Y657X, which truncates the cSH2 domain (Huang-Doran *et al.*, 2016; Kwok *et al.*, 2020), and E489K which, atypically, lies in the inter-SH2 domain where most cancer, overgrowth and APDS2-associated mutations lie (Thauvin-Robinet *et al.*, 2013). PIK3R1 E489K-containing primary cells were previously suggested to show basal hyperactivation (Thauvin-Robinet *et al.*, 2013).

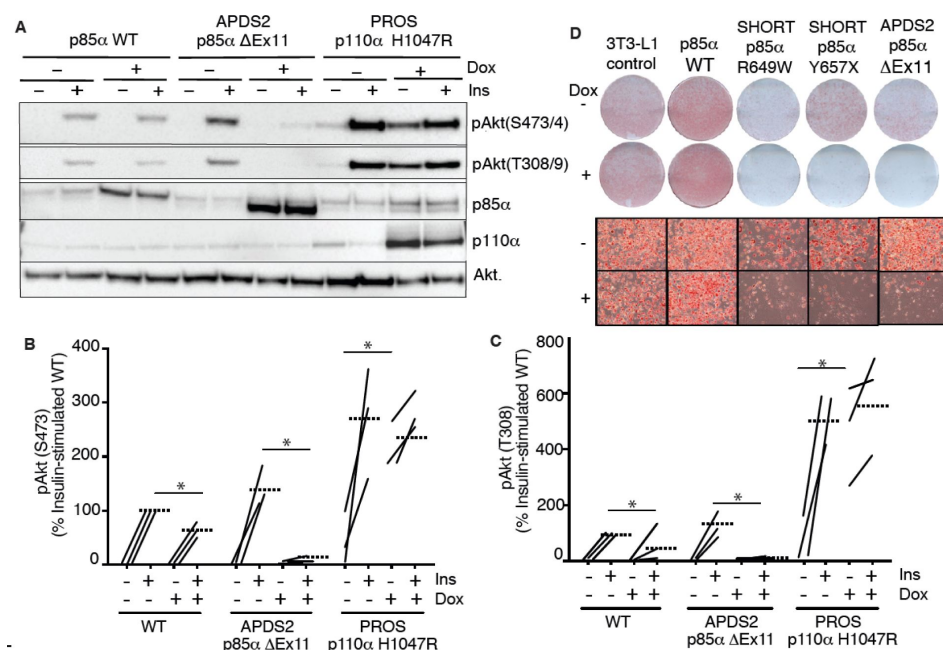


Figure 2.

Blunted insulin signaling in 3T3-L1 preadipocyte models of APDS2 and SHORT syndrome.

Immunoblotting of Akt, Akt phosphorylated at threonine 308 (T308) or serine 473 (S473), p85α, and p110α and are shown with and without stimulation with 100 nM insulin (Ins) for 10 minutes. Cells were treated with doxycycline (Dox) 1 μg/mL for 72 hours prior to insulin stimulation as indicated. (A) One immunoblot representing 3 experiments is shown. (B)-(C) Quantification of immunoblot bands from all 3 independent experiments shown for phosphoAkt-S473 and phosphoAkt-T308 respectively. Paired datapoints +/- insulin are shown, and dotted lines mark means. Asterisks indicate a significant difference. More detailed statistical analysis including 95% confidence intervals for the paired mean differences for these comparisons are shown in Figure EV3. (D) Staining for neutral lipid with Oil Red O of 3T3-L1 cells at day 10 of adipocyte differentiation. Induction of transgene expression by 1 μg/mL doxycycline throughout differentiation is shown. Images of entire plates are shown above, with representative bright field microscopy images below.

Wild-type and SHORT syndrome mutant holoenzymes were successfully purified for *in vitro* assay, but despite multiple attempts, Δ Ex11 holoenzyme could only be made in small amounts under identical conditions, and moreover was unstable on storage, precluding further study. In keeping with previous reports (Chudasama *et al.*, 2013 [↗](#); Dornan *et al.*, 2017 [↗](#); Dornan *et al.*, 2020 [↗](#)), p85 α R649W showed severely reduced phosphotyrosine-stimulated activity in complex with p110 α , with highly significant but lesser loss of function seen for Y657X and E489K (Fig 3A [↗](#), Fig 3 supplement 1A [↗](#)). No increase in basal activity was seen for any variant.

We also assessed whether any of the SHORT syndrome variants affect function of PI3K β and found that all variants impaired phosphotyrosine-stimulated activity of PI3K β . Again, this impairment was less for p85 α Y657X than for R649W, and in this case only very mild for E489K (Fig 3B [↗](#), Fig 3 supplement 1B [↗](#)). For PI3K δ , p85 α R649W conferred severe loss of function, as for other isoforms (Fig 3C [↗](#), Fig 3 supplement 1C [↗](#)), suggesting that the absence of immunodeficiency in SHORT syndrome is not accounted for by selective inhibition of PI3K α function by causal mutations.

Binding of p110 α by mutant p85 α

Given the potent dominant negativity of APDS2-related p85 α Δ Ex11 in cells, and the instability of p85 α Δ Ex11-containing PI3K holoenzyme *in vitro*, we next used immunoprecipitation to assess binding of p110 α by this and other p85 α variants in cells. p85 α was easily detected in anti-p110 α immunoprecipitates from all cell lines at baseline (Fig 4 [↗](#)). No increase was seen on conditional overexpression of wild-type or R649W p85 α . Given the known instability of monomeric p110 α , this suggests that all p110 α is bound to p85 α before overexpression. Although wild-type endogenous p85 α may have been replaced by heterologously overexpressed p85 α in these cells, this could not be detected without a size difference of the variant from wild-type. For cells overexpressing the SHORT syndrome-associated p85 α Y657X, the truncated variant was strongly co-immunoprecipitated, accounting for nearly all of the p85 α signal in anti-p110 α immunoprecipitate. This demonstrates preserved binding of p110 α by mutant p85 α (Fig 4 [↗](#)). In sharp contrast, although truncated p85 α Δ Ex11 was easily detected in cell lysates before immunoprecipitation and in supernatant after immunoprecipitation (arrowed in Fig 4 [↗](#)), no truncated p85 α Δ Ex11 was seen in p110 α immunoprecipitates, and no change in p110 α expression was detected (Fig 4 [↗](#)). This suggests that this truncated APDS2 causal variant does not supplant endogenous, full-length p85 α binding to p110 α , despite overexpression. This argues against destabilisation of p110 α as the mechanism of the observed dominant negative activity.

Effect of PIK3R1 mutations on insulin-induced PI3K recruitment to IRS1/2

As APDS2 p85 α Δ Ex11 does not appear to displace wild-type p85 α from p110 α , despite strong overexpression, it is likely that there are high levels of the monomeric truncated p85 α in the cell. This may be important as it is p85 α that mediates recruitment of PI3K to activated tyrosine kinase receptors and their tyrosine phosphorylated substrates, including the insulin- receptor substrate proteins Irs1 and Irs2. Excess free regulatory subunits compete with heterodimeric PI3K holoenzyme (Ueki *et al.*, 2002 [↗](#)), raising the possibility that excess free, truncated APDS2 p85 α Δ Ex11 may exert its inhibitory action by competing with PI3K holoenzyme.

To assess this possibility, we again used the 3T3-L1 cellular model to determine whether overexpression of disease-causing p85 α variants impairs recruitment of p110 α to Irs1 and Irs2. Irs1 was immunoprecipitated with or without conditional p85 α overexpression and with or without insulin stimulation. Overexpression of wild-type p85 α had no effect on basal or insulin-induced p110 α recruitment to Irs1 (Fig 5A,B [↗](#), Fig 5 supplement 1A,B [↗](#)). In contrast, overexpression of either p85 α R649W or Y657X sharply reduced insulin-stimulated p110 α recruitment to Irs1 on insulin stimulation (Fig 5A,B [↗](#), Fig 5 supplement 1A,B [↗](#)). In keeping with the inability of the R649W cSH2 domain to bind phosphotyrosines, p85 α R649W was also not recruited to Irs1, while overexpression of Y657X increased stimulated but not basal p85 α

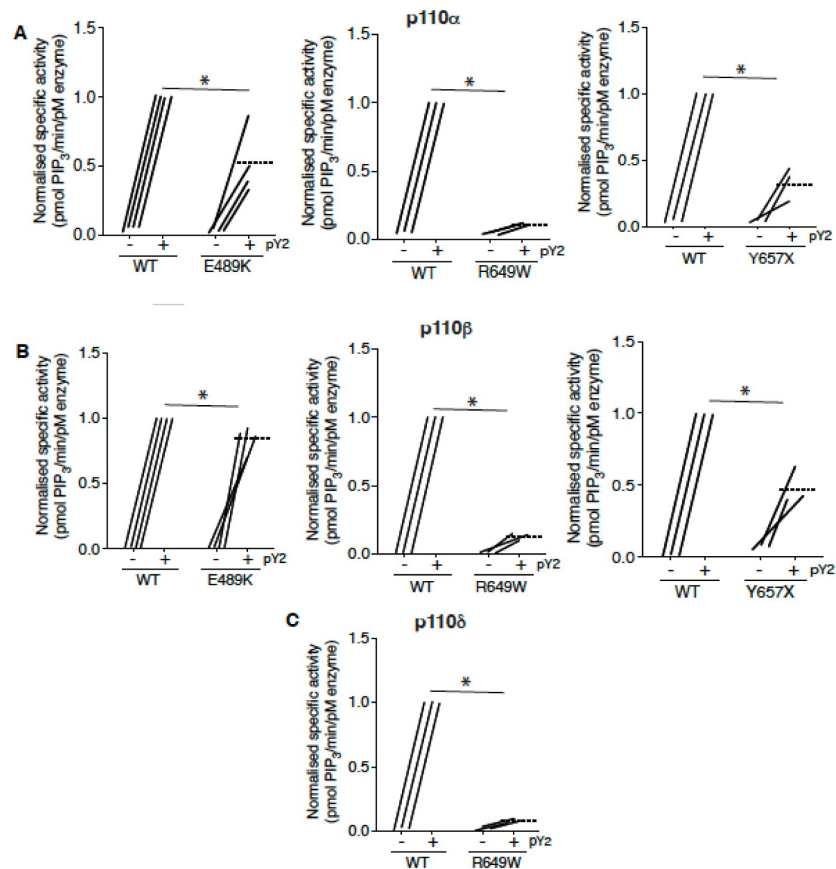


Figure 3.

SHORF syndrome p85α mutations impair phosphotyrosine-stimulated phosphoinositide 3-kinase activity. Lipid kinase activity of purified recombinant PI3K complexes generated using baculoviral expression in *Sf9* cells was measured using a modified fluorescence polarisation assay. WT p85α or p85α SHORF syndrome mutations, E489K, R649W or Y657X bound to either (A) p110α (B) p110β or (C) p110δ were assayed for basal and bisphosphotyrosine (pY2)-stimulated lipid kinase activity. Dotted lines mark means, and asterisks indicate a significant difference between the bisphosphotyrosine (pY2)-stimulated state for WT and comparator mutant p85α. More detailed statistical analysis including 95% confidence intervals for the paired mean differences for these comparisons are shown in Figure EV5.

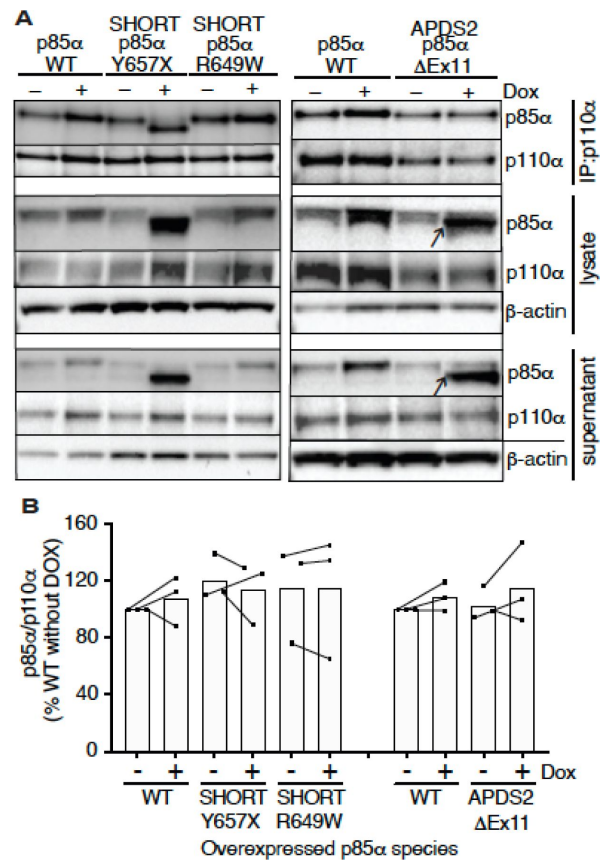


Figure 4.

Ability of pathogenic p85α variants to bind p110α, assessed by co-immunoprecipitation. Results of immunoblotting of anti-p110α immunoprecipitates from 3T3-L1 cells expressing wild-type, APDS2-associated or SHORT syndrome-associated mutant p85α under the control of doxycycline (Dox) are shown. (A) One representative immunoblot of immunoprecipitate, cell lysate prior to immunoprecipitation, and post immunoprecipitation supernatant is shown. (B) Quantification of immunoblot bands from immunoprecipitates from 3 independent experiments. Co-immunoprecipitated p85α is shown normalized to immunoprecipitated p110α from all 3 independent experiments. Datapoints from the same experiment +/- doxycycline are connected by lines. No significant differences were found among conditions.

recruitment (**Fig 5A,C**, **Fig 5 supplement 1C,D**). This suggests a pure defect in PI3K holoenzyme recruitment for R649W. For Y657X the signalling defect may have mixed mechanisms, with reduced activation by phosphotyrosine seen in *in vitro* studies coupled to increased abundance of monomeric mutant p85 α , leading to recruitment of non p110 α - bound mutant p85 α to Irs1.

In keeping with our finding of severely attenuated insulin signalling upon p85 α Δ Ex11 overexpression (**Fig 2**), overexpression of this mutant p85 α abolished p110 α recruitment to Irs1 (**Fig 5A,B**, **Fig 5 supplement 1A,B**). However, non p110 α -bound p85 α Δ Ex11 was strongly recruited to Irs1 even in the absence of insulin stimulation (**Fig 5A,C**, **Fig 5 supplement 1C,D**). This suggests that although p85 α Δ Ex11 does not effectively compete with wild-type p85 α for binding to p110 α , it has preserved or possibly enhanced ability to bind Irs1. This gives the mutant p85 α properties that render it a more potent endogenous inhibitor of PI3K signalling than free wild-type p85 α . Experiments conducted with immunoprecipitation of Irs2 instead of Irs1 yielded closely similar findings for all p85 α species (**Fig 6** and **Fig 6 supplement 1**).

Discussion

Murine genetic studies of *Pik3r1* have proved complicated to interpret, due in part to functional redundancy among PI3K regulatory subunits, and in part to unanticipated effects of perturbing PI3K subunit stoichiometry. This has left several important questions about *in vivo* functions of *Pik3r1* unresolved. Identification of a series of human genetic disorders caused by constitutional *PIK3R1* mutations over the past 10 years has given fresh impetus to the field and has been mechanistically illuminating.

Homozygous truncating *PIK3R1* mutations abolishing p85 α expression while preserving p55 α and p50 α produce agammaglobulinaemia (Conley *et al.*, 2012; Tang *et al.*, 2018). This resembles the immunodeficiency reported in *Pik3r1* knockout mice (Fruman *et al.*, 1999; Suzuki *et al.*, 1999) and suggests an essential, non-redundant function of p85 α in B cell development in humans. Thereafter, human genetics has provided more novel insights. *PIK3R1* mutations were identified in SHORT syndrome in 2013 (Chudasama *et al.*, 2013; Dymant *et al.*, 2013; Thauvin-Robinet *et al.*, 2013), nearly all in the C-terminal SH2 domain, which together with the N-terminal SH2 domain enables PI3K recruitment to activated RTKs (Rordorf-Nikolic *et al.*, 1995). SHORT syndrome features short stature and insulin resistance consistent with impaired ligand-induced p110 α action, a phenotype distinct from the enhanced insulin sensitivity produced by genetic ablation of one or more *Pik3r1* products in mice (Chen *et al.*, 2004; Fruman *et al.*, 2000; Mauvais-Jarvis *et al.*, 2002). Mice with SHORT syndrome mutations knocked in were generated after human findings, and faithfully reproduce the human phenotype (Kwok *et al.*, 2020; Solheim *et al.*, 2018; Winnay *et al.*, 2016), confirming that expression of a signalling-impaired *PIK3R1* has different consequences to *Pik3r1* knockout. No immunodeficiency has been described associated with classic SHORT syndrome mutations, however, suggesting a selective effect on PI3K α , but the basis of this selectivity has not previously been investigated.

In contrast to SHORT syndrome, mutations in the inter SH2 domain of *PIK3R1*, mostly leading to skipping of exon 11, were shown in 2014 to activate PI3K *in vitro* and to cause immunodeficiency (APDS2) (Deau *et al.*, 2014; Lucas *et al.*, 2014b) similar to that caused by activating mutations in p110 δ (APDS1) (Angulo *et al.*, 2013; Lucas *et al.*, 2014a). Neither overgrowth nor metabolic features of APDS2 have been described to suggest p110 α hyperactivation, and indeed short stature is common in APDS2 (Elkaim *et al.*, 2016; Jamee *et al.*, 2020; Maccari *et al.*, 2023; Olbrich *et al.*, 2016; Petrovski *et al.*, 2016), with a growing number of APDS2 patients described with features of SHORT syndrome (Bravo Garcia- Morato *et al.*, 2017; Maccari *et al.*, 2023; Nguyen *et al.*

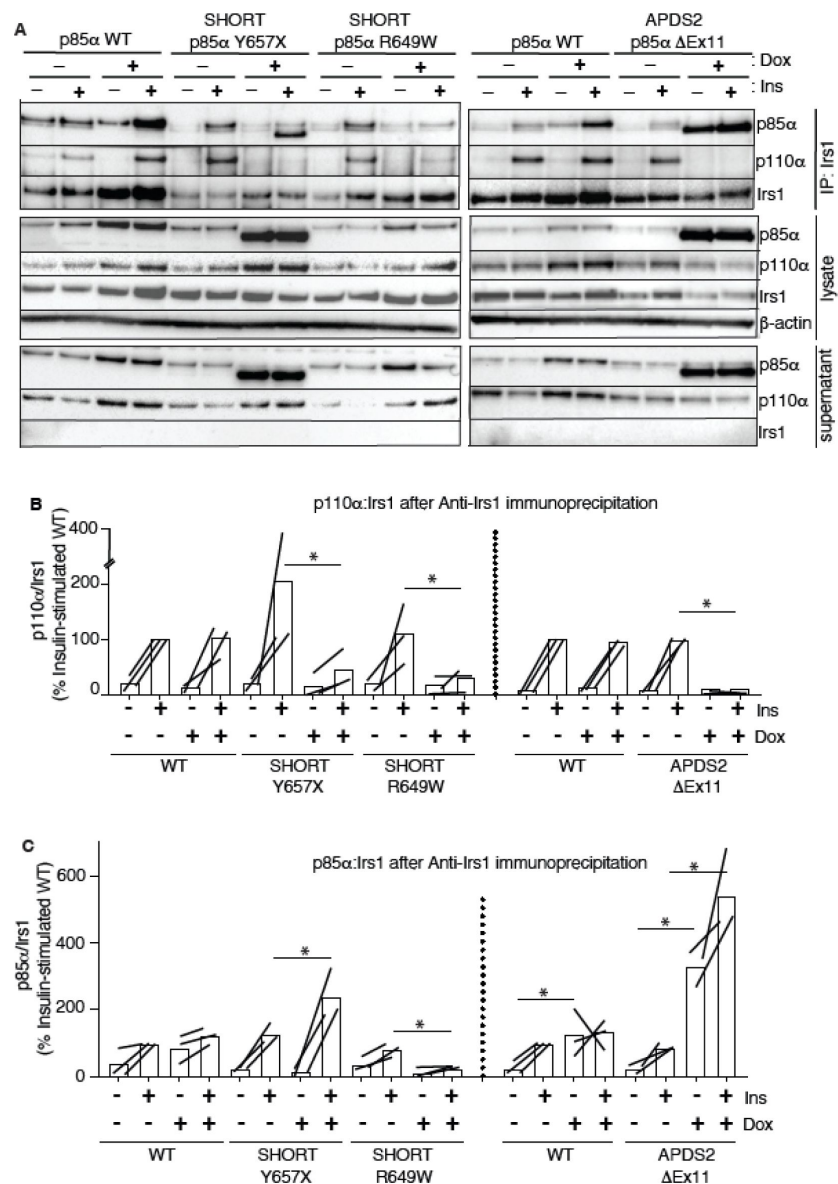


Figure 5.

Attenuated insulin-induced association of p110α with Irs1 in the presence of APDS2 and SHORT syndrome mutant p85α. Results of immunoblotting of anti-Irs1 immunoprecipitates from 3T3-L1 cells expressing wild-type, APDS2-associated or SHORT syndrome-associated mutant p85α under the control of doxycycline (Dox) are shown. Treatment with 100nM insulin (Ins) is indicated. (A) One representative immunoblot of immunoprecipitate, cell lysate prior to immunoprecipitation, and post immunoprecipitation supernatant is shown. Two separate sets of gels, including independent wild-type controls are shown on left and right. (B-C) Quantification of immunoblot bands from immunoprecipitates from 3 independent experiments. Immunoprecipitated p110α is shown normalized to immunoprecipitated Irs1 from all 3 independent experiments in (B), and immunoprecipitated p85α similarly in (C). Datapoints from the same experiment +/- insulin are connected by lines. Asterisks indicate significant differences induced by transgene overexpression (i.e. plus versus minus doxycycline). More detailed statistical analysis including 95% confidence intervals for the paired mean differences for these comparisons are shown in Figure EV6.

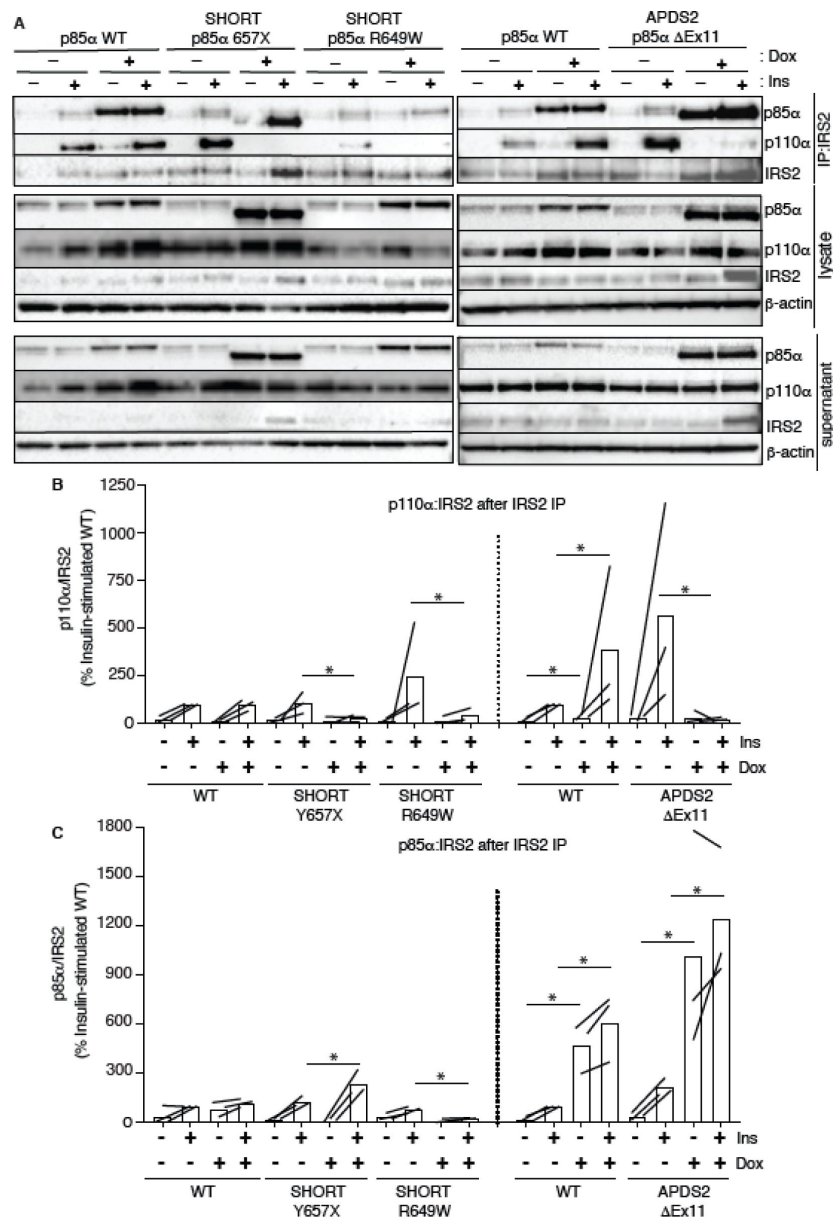


Figure 6.

Attenuated insulin-induced association of p110α with Irs2 in the presence of APDS2 and SHORT syndrome mutant p85α. Results of immunoblotting of anti-Irs2 immunoprecipitates from 3T3-L1 cells expressing wild-type, APDS2-associated or SHORT syndrome-associated mutant p85α under the control of doxycycline (Dox) are shown. Treatment with 100nM insulin (Ins) is indicated. (A) One representative immunoblot of immunoprecipitate, cell lysate prior to immunoprecipitation, and post immunoprecipitation supernatant is shown. Two separate sets of gels, including independent wild-type controls are shown on left and right. (B-C) Quantification of immunoblot bands from immunoprecipitates from 3 independent experiments. Immunoprecipitated p110α is shown normalized to immunoprecipitated Irs2 from all 3 independent experiments in (B), and immunoprecipitated p85α similarly in (C). Datapoints from the same experiment +/- insulin are connected by lines.

et al., 2023 [\[link\]](#); Petrovski *et al.*, 2016 [\[link\]](#); Ramirez *et al.*, 2020 [\[link\]](#); Szczawinska-Poplonyk *et al.*, 2022 [\[link\]](#)). Moreover mice with the common APDS2 causal *Pik3r1* variant knocked in show impaired growth and *in utero* survival, unlike APDS2 murine models (Nguyen *et al.*, 2023 [\[link\]](#)).

Thus, study of distinct PIK3R1-related syndromes shows that impaired signalling caused by PIK3R1 mutations produces phenotypes related to impaired PI3K α function, without affecting PI3K δ . Conversely, mutations of PIK3R1 that produce phenotypes related to increased PI3K δ signalling do not increase PI3K α signalling. Indeed, they surprisingly have features characteristic of impaired PI3K α function.

Lack of overgrowth in APDS2 has been attributed to greater ability of APDS2 PIK3R1 variants to activate p110 δ than p110 α (Dornan *et al.*, 2017 [\[link\]](#)), but the co-occurrence of gain- and loss- of function phenotypes has not been explained to date. Our findings suggest that the explanation may lie in competing effects of APDS2 PIK3R1 variants to activate PI3K δ on one hand, as previously shown (Angulo *et al.*, 2013 [\[link\]](#); Dornan *et al.*, 2017 [\[link\]](#); Lucas *et al.*, 2014b [\[link\]](#)), while on the other hand interfering with PI3K α through the dominant negative effect of non p110 α -bound mutant p85 α .

The low expression of truncated p85 α Δ Ex11 we described in dermal fibroblasts is similar to observations made in lymphocytes, however, in lymphocytes this is associated with increased basal AKT phosphorylation (Deau *et al.*, 2014 [\[link\]](#); Lucas *et al.*, 2014b [\[link\]](#)) that is abolished by p110 δ inhibition (Lucas *et al.*, 2014b [\[link\]](#)). P110 δ is also expressed in fibroblasts, but protein levels were reduced in the cells studied, consistent with the previously reported observation that when expressed in insect cells the PI3K holoenzyme containing p85 α Δ Ex11 has a low yield and is unstable (Dornan *et al.*, 2017 [\[link\]](#)). We speculate that in cells with low endogenous p110 δ protein expression, the destabilising effect of mutant PIK3R1 predominates over its activating effect.

The balance between expression and signalling may be a fine one, however, as transient overexpression of FLAG-tagged p85 α Δ Ex11 did increase AKT phosphorylation in 3T3 fibroblasts in a previous study, although expression of p110 α , β and δ was not determined (Deau *et al.*, 2014 [\[link\]](#)). We find, in contrast, that overexpression of untagged p85 α Δ Ex11 has a strong inhibitory effect on insulin signalling, and we were unable to identify a window of overexpression where increased AKT phosphorylation could be observed. We further demonstrated that most or all of the mutant p85 α expressed is not bound to p110 α , while unbound mutant p85 α still binds to Irs1/2, effectively competing with PI3K holoenzyme. The observation that p85 α Δ Ex11 can associate with Irs1/2 is in agreement with reports that p85 α Δ Ex11-containing PI3K α and PI3K δ can be stimulated by pY2-peptides (Dornan *et al.*, 2017 [\[link\]](#); Dornan *et al.*, 2020 [\[link\]](#)) and that p85 α Δ Ex11 is recruited to tyrosine-phosphorylated LAT in T cells (Lucas *et al.*, 2014b [\[link\]](#)). The competition we suggest between unbound mutant p85 α and PI3K holoenzyme for binding to the activated RTKs is in keeping with longstanding evidence that free p85 downregulates PI3K signalling through the same competitive mechanism (Thorpe *et al.*, 2017 [\[link\]](#); Ueki *et al.*, 2002 [\[link\]](#)), with a single study suggesting in addition that overexpression of tagged p85 α leads to Irs1 sequestration with free p85 α in cytosolic foci where PI(3,4,5)P3 production does not occur (Luo *et al.*, 2005 [\[link\]](#)).

The current study has limitations. We have studied primary cells from only a single APDS2 patient, and in the 3T3-L1 cell model, we did not determine whether p110 δ protein could be detected. If not, this could explain the lack of detectable AKT phosphorylation with induction of *Pik3r1* Δ Ex11. Our study moreover raises further questions. Full length p85 α subunits can homodimerize (Cheung *et al.*, 2015 [\[link\]](#); Harpur *et al.*, 1999 [\[link\]](#); LoPiccolo *et al.*, 2015 [\[link\]](#)), and it is speculated that homodimers may outcompete p85 α /p110 heterodimers for binding to activated Irs1 due to configuration of the four SH2 domains. In the 3T3-L1 preadipocyte APDS2 models, p85 α Δ Ex11 expression was high despite impaired p110 α association, but whether this is composed of

monomeric or homodimeric p85 α , and whether mutant p85 α expression leads to sequestration of Irs1 remote from the insulin receptor, as previously suggested for tagged wild-type p85 α , is undetermined.

In summary, it is already established that: A. genetic activation of PIK3CD causes immunodeficiency without disordered growth, while B. inhibition of PIK3R1 recruitment to RTKs and their substrates impairs growth and insulin action, without immunodeficiency, despite all catalytic subunits being affected and C. loss of p85 α alone causes immunodeficiency. The current study, coupled with prior reports, suggests that the common APDS2 mutation in PIK3R1 has mixed consequences, producing greater hyperactivation of p110 δ than p110 α , based on subtle differences in the inhibitory interactions of regulatory and catalytic subunits, while also destabilising PI3K holoenzyme and exerting dominant negative activity on wild-type PIK3R1 function. We suggest that these competing activating and inhibitory consequences are finely balanced, potentially differing among tissues, leading to mixed clinical profiles of gain- and loss-of function features.

Materials and Methods

Ethics

The patient studied gave written informed consent for NIH IRB-approved research protocols 06-I-0015 and 09-I-0133 and was reported previously (Lucas *et al.*, 2014b [↗](#)). Dermal fibroblasts were isolated from a skin punch biopsy and cultured as previously described (Huang-Doran *et al.*, 2016 [↗](#)).

Baculovirus Generation

p85 α point mutation expression constructs were created by site-directed mutagenesis of a pACEBac1_Homo sapiens p85 α plasmid using a QuikChange II XL Site-Directed Mutagenesis Kit. In-Fusion cloning was used to generate pACEBac1_p85 α Δ Ex11. In brief, pACEBac1_p85 α backbone was digested (BamHI/NotI) and purified using Zymoclean Gel DNA Recovery Kit, while cDNA insert was purified using NucleoSpin® Clean-up Kit (Takara). Insert generation was performed by High-Fidelity PCR using primers designed to yield fragments containing exons 1-10 or 12-16 with the necessary overlap, with inserts verified by electrophoresis and purified using NucleoSpin® Clean-up kit. In-Fusion reactions were performed according to manufacturer's guideline (Takara) using 100ng each of linearised plasmid and both inserts in 10 μ L. Products were transformed into Stellar Competent cells (Takara). Purified plasmid inserts were sequenced and verified by restriction enzyme digest, exploiting loss of a DpnI site within excised exon 11. pACEBac1_p85 α and pFastBac™HT B plasmids encoding N terminally tandem His-tagged p110 subunits were used to generate bacmid DNA. MAX Efficiency DH10Bac™ Competent or EMBacY cells (ThermoFisher) were transformed with 40ng plasmid DNA and turbid cultures plated onto agar containing 50 μ g/mL kanamycin, 10 μ g/mL gentamicin, 10 μ g/mL tetracycline, 100 μ g/mL X-Gal and 40 μ g/mL IPTG. Single colonies were picked, expanded and purified at 2 days and bacmid concentrations quantified by Nanodrop 1000 Spectrophotometer (Thermo Scientific). 2-4 μ g bacmid was transfected into Sf9 cells in 6 well plates using Insect-XPRESS (Promega E2311). Cells were incubated at 2°C for 5 days and bacmid YFP expression assessed by Leica DM IL LED Fluo microscope using a green fluorescent protein filter cube. Pooled virus from 2 wells of supernatant (P1 stock) was used to produce high titre P2 stock by transfection of 1.5x10⁶ Sf9 cells/mL (ThermoFisher) in 450mL using Insect-XPRESS with L-Glutamine (Lonza LZBE12-730Q). For catalytic subunits, 1-2mL P2 stock was used for a further round of Sf9 transfection and expansion to generate P3 stock.

Purification of PI3K holoenzymes

1.5–2 L of 1.5×10^6 Sf9 cells/mL were co-infected with 18 mL P3 catalytic subunit and 4 mL P2 regulatory subunit baculovirus. Non-infected Sf9 cells were negative controls, and a prior protein preparation of 320 kDa was the positive control. Cells were cultured at 27°C, harvested 48 h post-infection, pelleted, washed and stored at –80°C. Pellets were later lysed by sonication in buffer containing 20 mM TrisHCl (pH 8.0), 300 mM NaCl, 20 mM imidazole (pH 8.0) and 1 mM TCEP (pH 7.0) at 4°C. Universal Nuclease (ThermoFisher) was added to lysates before ultracentrifugation at 35,000 rpm for 35 min at 4°C. p85 α /p110 heterodimers were pulled down via 6 tandem p110 N-terminal His tags, preventing purification of monomeric p85 α , using tandem Ni²⁺/HisTrap Fast Flow columns (GE Healthcare) (equilibration buffer 20 mM TrisHCl (pH 8.0), 100 mM NaCl, 20 mM imidazole and 1 mM TCEP (pH 7.0); elution buffer 20 mM TrisHCl (pH 8.0), 100 mM NaCl, 200 mM imidazole and 1 mM TCEP (pH 7.0)). Further purification utilised a heparin HiTrap Q HP column (GE Healthcare) equilibrated with 20 mM TrisHCl (pH 8.0) and 1 mM TCEP (pH 7.0), with proteins eluted in 20 mM TrisHCl (pH 8.0), 1 M NaCl and 1 mM TCEP (pH 7.0). Eluted fractions were concentrated to ≥ 1 mg/mL using Millipore Amicon Ultra-15 Centrifugal Filter Units with Ultracel-50 membrane, and 1 mL concentrated fractions were gel filtered on Superdex 200 16/60 columns (GE Healthcare) equilibrated in 20 mM HEPES (pH 7.5), 100 mM NaCl and 2 mM TCEP (pH 7.0). The p110 6-His tag was retained for proteins used in functional analyses. KTA Protein Purification Systems (GE Healthcare) and UNICORN Control Software version 5.11 (GE Healthcare) were used for all purifications. Purity of eluted complexes was verified by SDS-PAGE, and purified proteins were quantified using a Nanodrop 1000 Spectrophotometer (ThermoFisher). Protein concentration was calculated using the molecular extinction coefficient (assuming full cysteine reduction) determined by heterodimer sequence input to the ProtParam tool (ExPASy). Purified proteins were stored at –80°C in single-use aliquots.

Total injection volume onto gel filtration columns was kept at 1 mL. Yields of PI3K complexes were determined using the area under the curve (280 nm MAU per mL eluted protein) from gel filtration chromatograms, and normalised to the volume of Sf9 cells phosphatidylserine and porcine brain PI(4,5)P₂ were mixed to generate a preparation containing 5/10/15/45/20/5% of each lipid, with a total lipid concentration of 5 mg/mL and final PI(4,5)P₂ concentration 250 μ g/mL. Lipid preparations were dessicated under argon and then in a vacuum desiccator. Lipids were rehydrated in buffer containing 20 mM HEPES (pH 7.5), 100 mM KCl and 1 mM EGTA (pH 8.0) using vortexing for 3 min, waterbath sonication for 15 min, and 10 cycles between liquid nitrogen and a 43°C water bath. Unilamellar vesicles were generated by extrusion through polycarbonate filters with 100 nm pores, using a glass-tight syringe. Single use aliquots were stored at –80 °C.

Fluorescence Polarisation Assay

PI(3,4,5)P₃ production was measured by modified PI3-Kinase activity fluorescence polarisation assay (Echelon Biosciences, Salt Lake City, UT, USA). 10 μ L reactions in 384-well black microtitre plates used 1 mM liposomes containing 50 μ M PI(4,5)P₂, optimised concentrations of purified PI3K proteins, 100 μ M ATP, 2 mM MgCl₂, with or without 1 μ M tyrosine bisphosphorylated peptide derived from PDGFR β (“pY2”; Cambridge peptides). Reactions were quenched with 5 μ L of solution containing 20 mM HEPES (pH 7.5), 150 mM NaCl, 30 mM EDTA (pH 7.4) and 400 nM GST–Grp1-PH, followed by addition of 5 μ L 40 nM TAMRA Red Fluorescent Probe in HNT buffer. Identity of the lipid group coupled to the TAMRA probe was not disclosed by Echelon Biosciences, but as Grp1 recognises specific lipid head groups in competition with PI(3,4,5)P₃, it is likely to be a lipid with a similar head group such as inositol 1,3,4,5-tetrakisphosphate. After 1 h equilibration, fluorescence polarisation was measured using a PHERAstar spectrofluorometer (BMG Labtech, Ortenberg, Germany) with the FP/540-20/590-20/590-20 optical module. The concentrations of wild-type protein complexes used in the assay were: 20–40 nM p85 α /p110 α , 80–160 nM p85 α /p110 β and 80–160 nM p85 α /p110 δ for basal activities, and 1–2 nM p85 α /p110 α , 2–4 nM p85 α /p110 β and 2–6 nM p85 α /p110 δ for pY2-stimulated activities. Concentrations of thawed protein aliquots were assessed as 280 nm absorbance by Nanodrop 1000 Spectrophotometer, and samples of preparations used in

assays were resolved by SDS- PAGE to confirm equal amounts of each complex. Three technical replicates were used per assay on at least three different occasions. Relevant wild-type p85 α /p110 catalytic isoforms were included in each assay as controls. Other controls were 1. Stop mix + probe without lipids, to assess maximum fluorescence polarisation. 2. Stop mix + probe with lipids, to determine the effect of lipids on maximum fluorescence polarisation. 3. Probe without stop mix or lipids to determine minimum fluorescence polarisation and 4. Probe with lipids but no stop mix to determine the effect of lipids on minimum fluorescence polarisation. Standard curves of diC8-PI(3,4,5)P3 in the presence of liposomes were included in every experiment and used to infer PIP3 generated by purified PI3K. Graphpad Prism v6.0 was used to generate sigmoidal standard curves by plotting log transformed diC8-PI(3,4,5)P3 concentrations against fluorescence polarisation. Standard curves for each experiment provided the linear range within which PIP3 could be accurately determined. Preliminary studies found only trivial differences in measured fluorescence polarisation in the presence of pY2, which was thus omitted from subsequent standard curves. Quadruplicate interpolated PIP3 concentrations were averaged and normalised by reaction time (30 min) and enzyme concentration in pM to yield PI3K activity in pmol PIP3/min/pM enzyme. Specific activities were then further normalised to activity of p85 α wild-type controls in the presence of pY2 in each experiment.

Generation of 3T3-L1 cells conditionally expressing p85 α or p110 α

3T3-L1 preadipocytes from Zenbio were cultured, differentiated and stained for neutral lipid with Oil Red O as previously described(Huang-Doran *et al*, 2011 [↗](#)). Generation of 3T3-L1 sublines inducibly expressing WT, Y657X, or R649W p85 α was also previously reported(Huang-Doran *et al.*, 2016 [↗](#); Hussain *et al*, 2011 [↗](#)). The 3T3-L1 lines inducibly expressing p85 α Δ Ex11 or p110 α H1047R were generated using essentially the same procedure, starting with In-Fusion subcloning of the p85 α Δ Ex11 cDNA insert from the pACEBac1 p85 α Δ Ex11 plasmid described above, and of a PIK3CA H1047R cDNA insert derived by PCR from a pcDNA5.FRT.TetON_FLAG_PIK3CA-H1047R plasmid, into the pEN_Tmcs entry vector. Transgene expression was induced with 1 μ g/mL doxycycline for 72 hours.

Insulin signalling studies

3T3-L1 cells serum starved in DMEM containing 0.5 % BSA for 16h were stimulated with 100nM Actrapid insulin (Novo Nordisk) for 10min. Monolayers were snap frozen in liquid N2 and stored at -80°C. 200-600 μ L co-IP lysis (20mM HEPES, 150mM NaCl, 1.5mM MgCl2, 10% (v/v) glycerol, 1% (v/v) Triton X-100, 1mM EGTA pH 7.4, 1mM PMSF, 2mM activated sodium orthovanadate, Complete Mini EDTA-free protease inhibitor cocktail and PhosSTOP, in Milli-Q Ultrapure water (Millipore)) or RIPA buffer (50mM Tris HCl pH 8.0, 150mM NaCl, 1% NP-40, 0.5% sodium deoxycholate, 0.1% SDS with added Complete Mini EDTA-free protease inhibitor cocktail and PhosSTOP, in Milli-Q Ultrapure Water (Millipore)) was added to frozen cells before scraping of lysate into pre-chilled tubes, incubation on ice for 30min and clearing by centrifugation. Protein was quantified using the Bio-Rad DC assay.

Lysates were mixed with NuPAGE® SDS Loading buffer supplemented with 5% L-mercaptoethanol and boiled before SDS-PAGE. For co-immunoprecipitation (Co-IP) lysates (150-300 μ g protein) were incubated in 500 μ L Co-IP buffer with immunoprecipitating antibody overnight at 4°C. 10 μ g lysate mixed with NuPAGE® SDS Loading buffer and 5% L-mercaptoethanol was stored to represent Co-IP input. Co-IP samples were incubated for 2h with 1.5mg PBST-washed Protein G Dynabeads® (Life Technologies) at 4°C before centrifugation and bead removal using a DynaMag™-2 Magnet (Invitrogen). Supernatants mixed 1:1 with Co-IP elution buffer (2X NuPAGE® LDS, 100 mM NuPAGE® Sample Reducing Agent, in Co-IP lysis buffer) were boiled for 10 min. Beads were washed 5 times in 50 μ L Co-IP lysis buffer, before protein elution with 25 μ L Co-IP Elution buffer and boiling for 10min. Input, supernatant and co-IP samples underwent SDS-PAGE using

NuPAGE® 4- 12% gradient Bis-Tris gels in NuPAGE® MOPS SDS Running Buffer. Transfer to nitrocellulose was performed using the iBlot™ Dry Blotting System (Invitrogen), with preincubation in Equilibration Transfer Buffer for proteins above 150kDa.

For immunoblotting, blocked membranes were incubated overnight at 4°C in primary antibody, washed and incubated with HRP-linked anti-rabbit or anti-mouse IgG secondary antibody diluted 1:5000 in blocking buffer. Proteins were visualized using the ChemiDoc™ MP System (Bio-Rad) and band intensities quantified using Image Lab software (Bio-Rad).

Statistical Analysis

For quantitative data, all biological replicates (i.e. results of independent experiments on different days) are represented in figures, with paired points (e.g. with/without insulin in the same experiment) connected by lines. For fluorescence polarisation assays, each biological replicate shown is the mean of 3 technical replicates. Sample size for individual experiments was not pre-determined. To avoid the pitfalls of dichotomous significance testing on low- throughput biological datasets, we used estimation statistics (Data Analysis using Bootstrap- Coupled ESTimation) with default settings (5000 resamples, BCa bootstrap)([Ho et al, 2019](#)). This focuses on effect sizes and derives confidence intervals derived from bootstrapping for differences in means; the small (but typical) sample size of the experiments analysed limits reliable bootstrapping, but it offers additional confidence to the consistent patterns seen in independent replicates. We have indicated significance in main text figures with an asterisk, and show the 95% confidence intervals for mean differences in extended view figures.

All antibodies used are listed in **Table 1** [↗](#).

Antibody	Company	Catalogue No.	Species	Dilution used (immunoblot/IP)
p85 α	CST	4257	Rabbit	1:1000
Phospho-AKT/Akt (T308)	CST	9275	Rabbit	1:1000
Phospho-AKT/Akt (S473)	CST	9271	Rabbit	1:1000
AKT/Akt (1,2,3)	CST	2920	Mouse	1:1000
p110 α	CST	4249	Rabbit	1:1000/1:50
p110 δ	CST	34050	Rabbit	1:1000
Irs1	Millipore	06-248	Rabbit	1:500
Irs1	CST	2382	Rabbit	1:1000/1:50
Irs2 clone 9.52	Millipore	MABS15	Mouse	1:1000/1:50
β -actin	CST	4967	Rabbit	1:5000
Anti-rabbit IgG; HRP-linked	CST	7074	Goat	1:5000
Anti-mouse IgG; HRP-linked	CST	7076	Horse	1:5000

Table 1

Antibodies Used

Data Availability

This study includes no data deposited in external repositories. Uncut Western blot and other source images are available on the Open Science Framework project webpage: https://osf.io/eyc4h/?view_only=9af8645c8fe5408790ce19920d0084c6 .

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Author Contributions

PRT: conceptualisation, methodology, formal analysis, investigation, visualization, writing - original draft. **RK:** formal analysis, investigation, visualization. **OP:** supervision, methodology, investigation, writing - review & editing. **HCS:** resources, funding acquisition, writing - review & editing. **GVB:** supervision, methodology, writing - review & editing. **RLW:** supervision, methodology, investigation, funding acquisition, writing - review & editing. **RKS:** conceptualisation, supervision, methodology, funding acquisition, writing - original draft, writing - review & editing, funding acquisition.

Disclosure and competing interests statement

RKS has consulted for Novartis on clinical aspects of PIK3CA-related overgrowth, and for Alnylam, Amryt and AstraZeneca on clinical aspects of monogenic insulin resistance and lipodystrophy

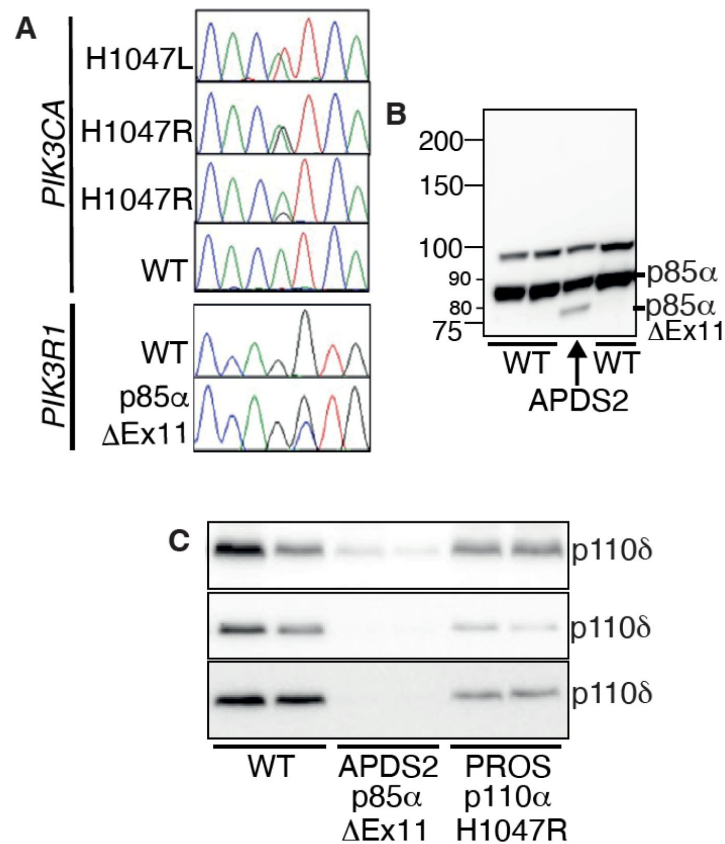


Figure 1 supplement 1.

Further characterisation of primary dermal fibroblasts studied. (A) Details of cDNA sequence for *PIK3CA* and *PIK3R1* from cells derived from healthy controls (WT), patients with APDS2 (p85a DEx11) or *PIK3CA*-related overgrowth syndrome (PROS), confirming expected expression of mutant alleles. (B) Higher magnification detail of immunoblot from wild-type and APDS2 fibroblasts showing truncated p85a DEx11. (C) All 3 immunoblot replicates for p110d blots quantified in [Figure 1](#), showing severely reduced p110d expression in the APDS2 cell line

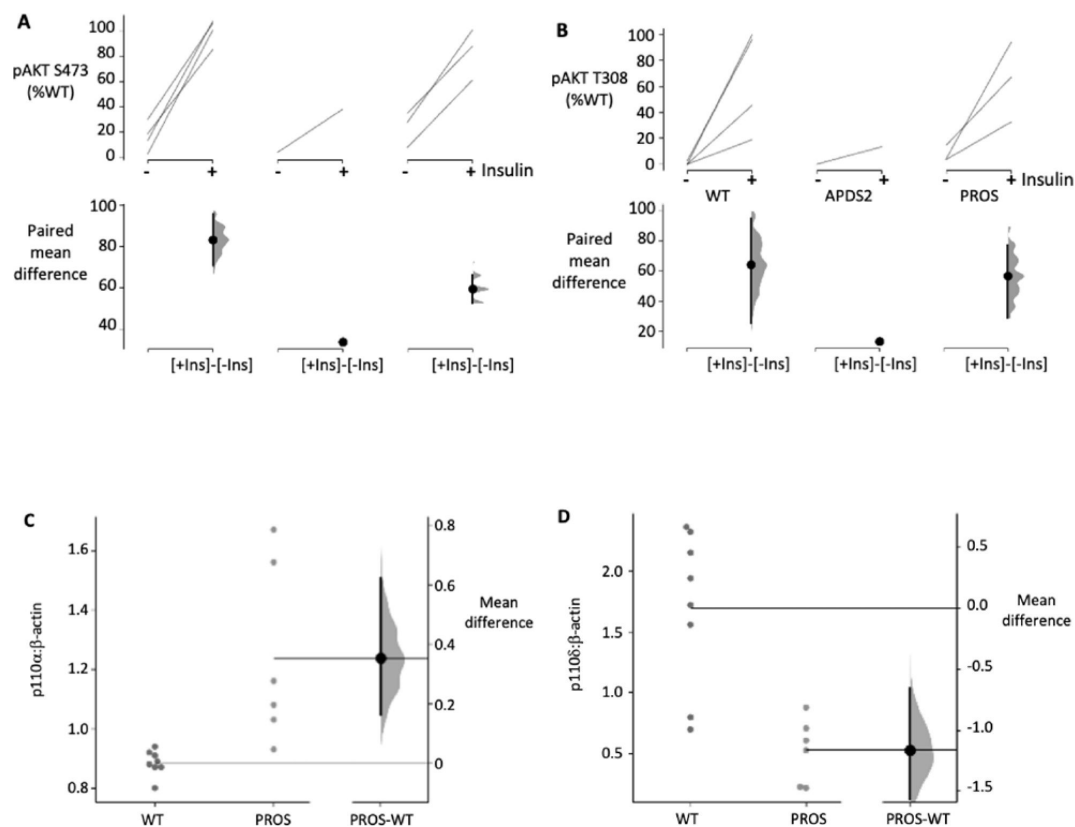


Figure 1 supplement 2.

Full statistical analysis of data presented in main

Figure 1 Analysis of insulin-induced increase in (A) AKT S473/4 and (B) T308/9 phosphorylation. The paired mean difference for 3-4 comparisons are shown in Cumming estimation plots. The raw data, as presented in **Figure 1**, are replotted on the upper axes with paired observations connected by a line. On the lower axes, paired mean differences are plotted as a bootstrap sampling distribution. Mean differences are depicted as dots; 95% confidence intervals are indicated by the ends of the vertical error bars. (C-D) Analysis of differences in (C) p110a and (D) p110d protein expression between healthy control cells and cells from PROS patients harbouring activating PIK3CA mutations. Mean differences are shown in Gardner-Altman estimation plots, with expression data plotted on the left axes and mean difference on floating axes on the right, again as a bootstrap sampling distribution with mean difference depicted as a dot and 95% confidence intervals by the ends of the vertical bar.

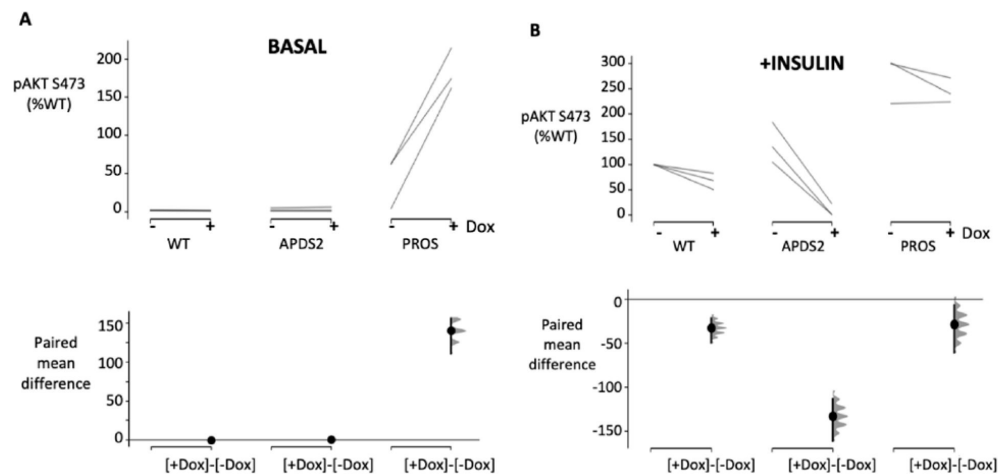


Figure 2 supplement 1.

Full statistical analysis of data presented in main

Figure 2 [↗](#). Analysis of the effects of doxycycline-induced expression of wild-type (WT) or Δ Ex11 (APDS2) p85 α , or of p110 α H1047R (PROS) on Akt S473/4 (A,B) and T308/9 (C,D) phosphorylation. Comparisons are made in both the basal, non-insulin stimulated state (A,C) and after stimulation with 10 nmol/L insulin (B,D). Paired mean differences for 3 comparisons are shown in Cumming estimation plots. Raw data, as presented in **Figure 2** [↗](#), are re-plotted on the upper axes with paired observations connected by a line. On the lower axes, paired mean differences are plotted as a bootstrap sampling distribution. Mean differences are depicted as dots; 95% confidence intervals are indicated by the ends of the vertical error bars.

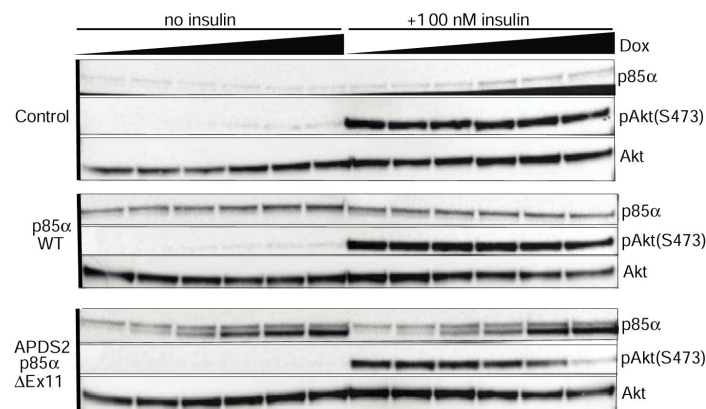


Figure 2 supplement 2.

The effect of graded expression of wild-type or disease-associated p85 α on 3T3-L1 preadipocytes. Immunoblots of p85 α , phosphoAkt (S473) and total Akt are shown for control 3T3-L1 cells and 3T3-L1 cells conditionally expressing wild-type (WT), or APDS2-associated mutant p85 α under the control of doxycycline (Dox), with and without 10 minutes of exposure to insulin as indicated. The filled black triangles indicate increasing concentrations of doxycycline (from left to right: 0, 0.02, 0.03, 0.045, 0.065, or 0.1 μ g/mL). Exposure was for 72 hours in all cases. The truncated p85 α variant can be seen below the WT p85 α for the APDS2 Δ Ex11 mutant.

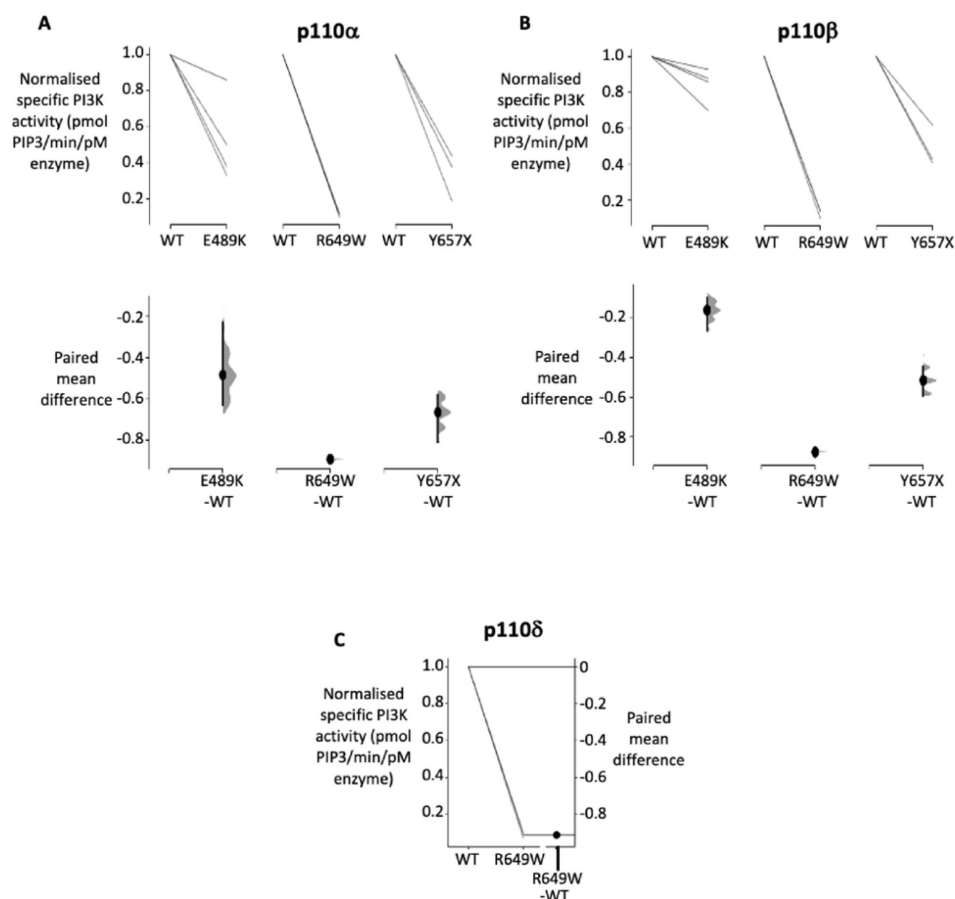


Figure 3 supplement 1.

Full statistical analysis of data presented in main

Figure 3. Analysis of fluorescence polarisation assay of phosphoinositide 3-kinase (PI3K) activity of *in vitro* synthesised wild-type (WT) or mutant (E489K, R649W or Y657X) p85 α . Results for p110 α - and p110 β -containing PI3K are shown in (A) and (B) respectively. All data were acquired in the presence of phosphotyrosine peptide. Paired mean differences for 3 comparisons are shown in Cumming estimation plots. Raw data, as presented in **Figure 3**, are re-plotted on the upper axes with paired observations connected by a line. On the lower axes, paired mean differences are plotted as a bootstrap sampling distribution. Mean differences are depicted as dots; 95% confidence intervals are indicated by the ends of the vertical error bars. Results for the R649W p85 α mutation only are shown with p110 δ in (C). In this case raw data are re-plotted on the left hand axes with paired observations connected by 3 nearly superimposed lines. On the right hand axes, paired mean differences are plotted as a bootstrap sampling distribution.

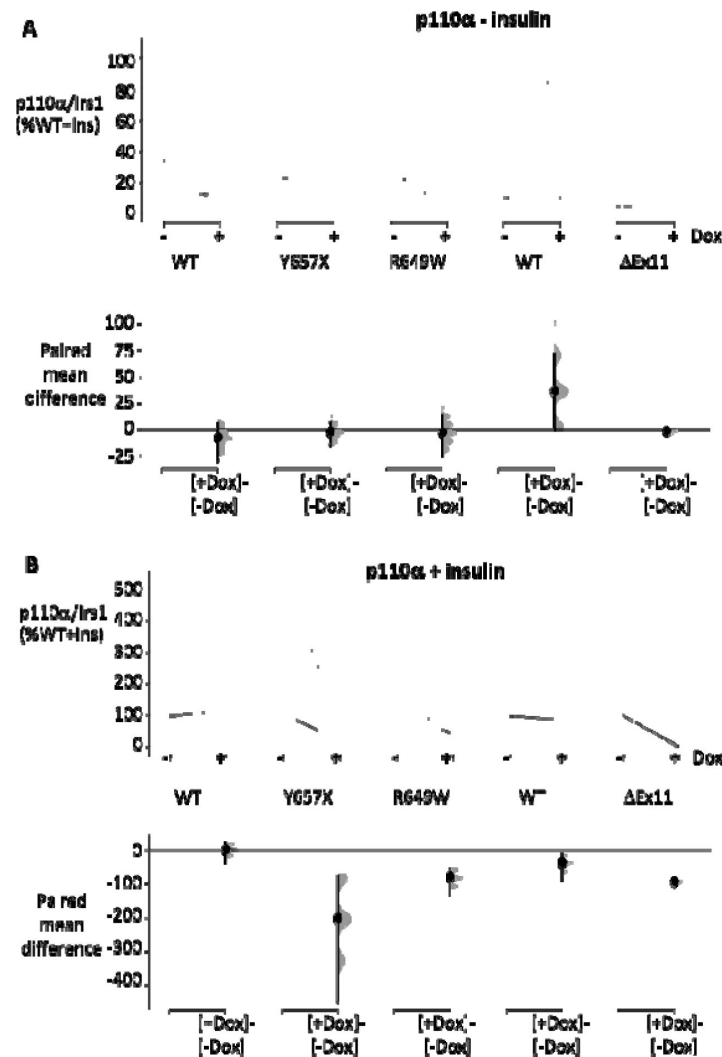


Figure EV5 supplement 1.

Full statistical analysis of data presented in main

Figure 5. Analysis of the effects of doxycycline (dox)-induced expression of wild-type (WT), Y657X, R649W or ΔEx11 p85α on association of p110α (A,B) and p85α (C,D) with Irs1. Results of co-immunoprecipitation with (B,D) and without exposure to 10 nmol/L insulin Comparisons are made in both the basal, non-insulin stimulated state (A,C) and after stimulation with 10 nmol/L insulin (B,D) are shown. Paired mean differences for 3 comparisons are shown in Cumming estimation plots. Raw data, as presented in **Figure 5**, are re-plotted on the upper axes with paired observations connected by a line. On the lower axes, paired mean differences are plotted as a bootstrap sampling distribution. Mean differences are depicted as dots; 95% confidence intervals are indicated by the ends of the vertical error bars.

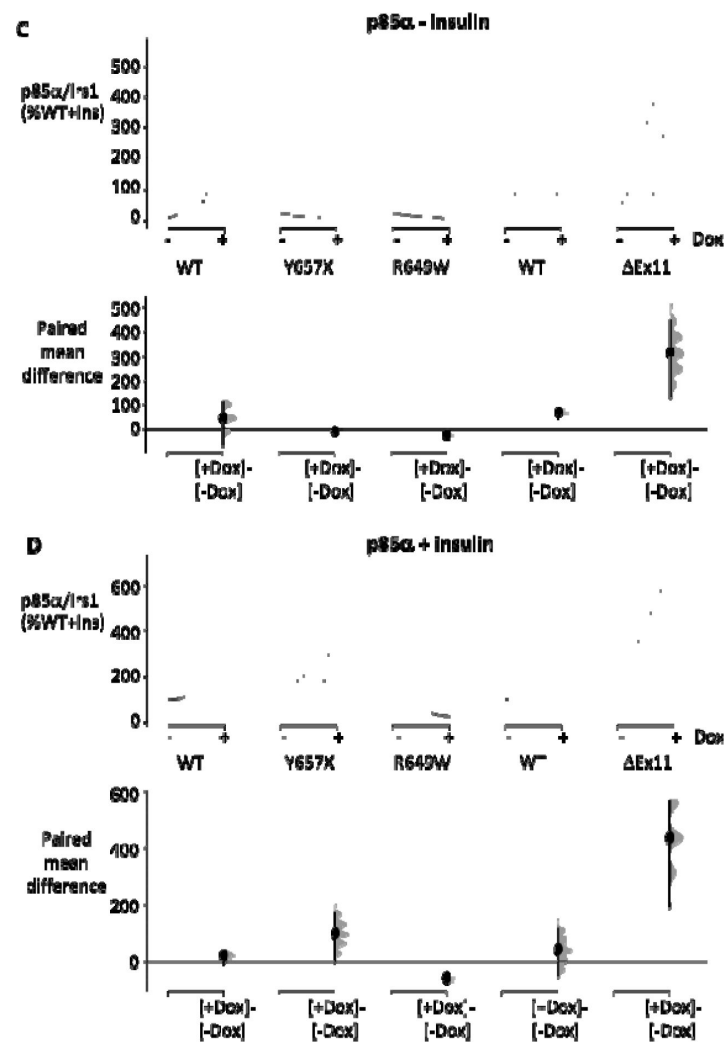


Figure EV5 supplement 1. (continued)

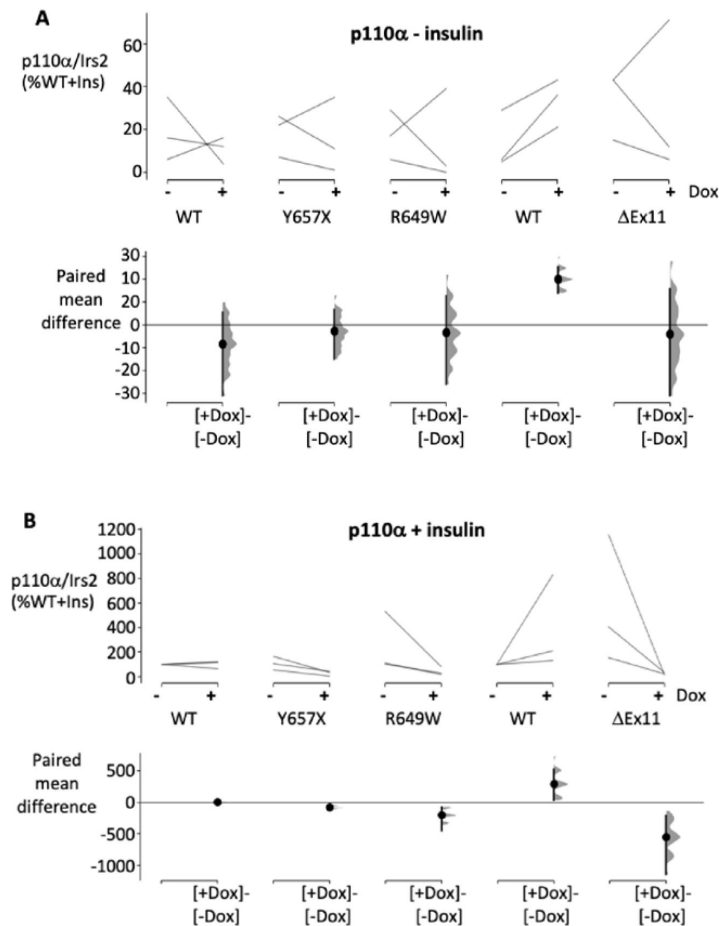


Figure 6 supplement 1.

Full statistical analysis of data presented in Figure EV7.

Analysis of the effects of doxycycline (dox)-induced expression of wild-type (WT), Y657X, R649W or ΔEx11 p85α on association of p110α (A,B) and p85α (C,D) with Irs2. Results of co-immunoprecipitation with (B,D) and without exposure to 10 nmol/L insulin. Comparisons are made in both the basal, non-insulin stimulated state (A,C) and after stimulation with 10 nmol/L insulin (B,D). Paired mean differences for 3 comparisons are shown in Cumming estimation plots. Raw data, as presented in Figure EV7, are re-plotted on the upper axes with paired observations connected by a line. On the lower axes, paired mean differences are plotted as a bootstrap sampling distribution. Mean differences are depicted as dots; 95% confidence intervals are indicated by the ends of the vertical error bars.

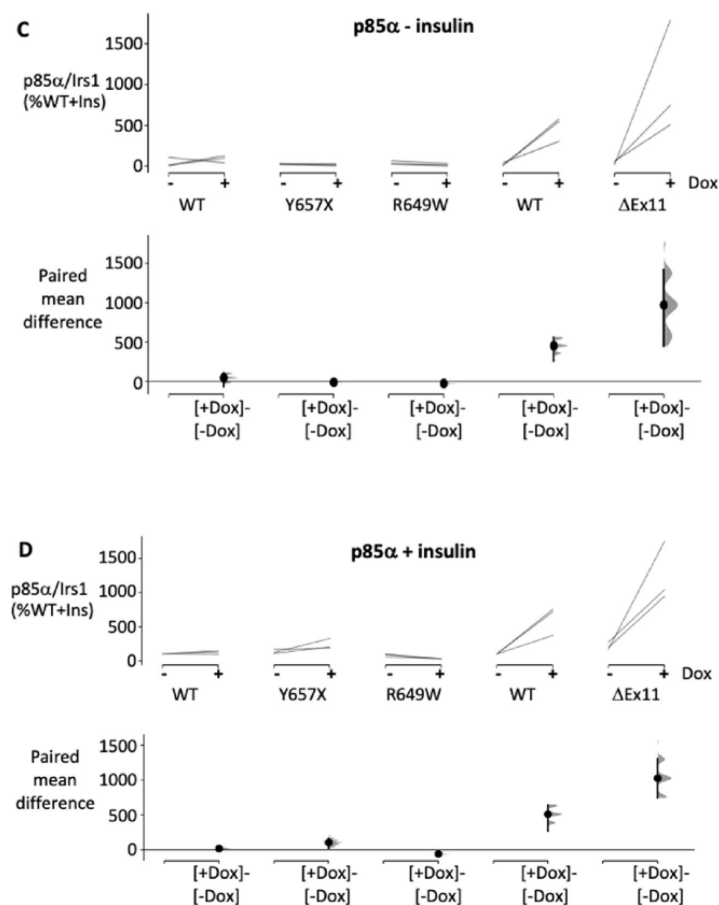


Figure 6 supplement 1. (continued)

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Reviewer #1 (Public Review):**Summary:**

This study provides convincing data showing that expression of the PIK3R1(delta Exon11) dominant negative mutation in Activated PI3K Delta Syndrome 1/2 (APDS1/2) patient-derived cells reduces AKT activation and p110δ protein levels. Using a 3T3-L1 model cell system, the authors show that overexpressed p85α delta Exon 11) displays reduced association with the p110α catalytic subunit but strongly interacts with Irs1/2. Overexpression of PIK3R1 dominant negative mutants inhibits AKT phosphorylation and reduces cellular differentiation of preadipocytes. The strength of this article is the clear results derived from Western blots analysis of cell signaling markers (e.g. pAKT1), and co-immunoprecipitation of PI3K holoenzyme complexes and associated regulatory factors (e.g. Irs1/2). The experimental design, interpretation, and quantification broadly support the authors' conclusions.

Strengths:

The authors analyze a variety of PIK3R1 mutants (i.e. delta Exon11, E489K, R649W, and Y657X), which reveals a range of phenotypes that support the proposed model for dominant

negative activity. The use of clonal cell lines with doxycycline-induced expression of the PIK3R1 mutants (ΔExon 11, R649W, and Y657X) provides convincing experimental data concerning the relationship between p85α mutant expression and AKT phosphorylation in vivo. The authors convincingly show that p85α delta Exon11, R649W, or Y657X is unable to associate with p110α but instead more strongly associates with Irs1/2 compared to wild type p85α. This helps explain why the authors were unable to purify the recombinant p110α/p85α delta Exon 11) heterodimeric complex from insect cells.

Weaknesses:

Future experimentation will be needed to reconcile the cell type specific differences (e.g. APDS2 patient-derived cells vs. the 3T3-L1 cell model system) in PIK3R1 mutant behavior reported by the authors. An unbiased proteomic study that broadly evaluates the cell signaling landscape could provide a more holistic understanding of the APDS2 and SHORT mutants compared to a candidate-based approach. Additional biochemical analysis of p110α/p85α delta Exon 11) complex is needed to explain why this mutant regulatory subunit does not strongly associate with the p110 catalytic subunit. It remains unclear why p85α delta Exon 11) expression reduces p110δ protein levels in APDS2 patient-derived dermal fibroblasts. This study would benefit from a more comprehensive biochemical analysis of the described p110α/p85α, p110β/p85α, and p110δ/p85α mutant protein complexes. The current limitation of this study to the use of a single endpoint assay to measure PI3K lipid kinase activity in the presence of a single regulatory input (i.e. RTK-derived pY peptide). A broader biochemical analysis of the mutant PI3K complexes across the canonical signaling landscape will be important for establishing how competition between wild-type and mutant regulatory subunits is regulated in different cell signaling pathways.

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Reviewer #2 (Public Review):

Summary:

Patsy R. Tomlinson et al; investigated the impact of different p85 alpha variants associated with SHORT syndrome or APDS2 on insulin-mediated signaling in dermal fibroblasts and preadipocytes. They find no evidence of hyperactive PI3K signalling monitored by pAKT in APDS2 patient-derived dermal fibroblast cells. In these cells p110 alpha protein levels were comparable to levels in control cells, however, the p110 delta protein levels were strongly reduced. Remarkably, the truncated APDS2-causal p85 alpha variant was less abundant in these cells than p85 alpha wildtype. Afterwards, they studied the impact of ectopically expressed p85 alpha variants on insulin-mediated PI3K signaling in 3T3-L1 preadipocytes. Interestingly they found that the truncated APDS2-causal p85 alpha variant impaired insulin-induced signaling. Using immunoprecipitation of p110 alpha they did not find truncated APDS2-causal p85 alpha variant in p110 alpha precipitates. Furthermore, by immunoprecipitating IRS1 and IRS2, they observed that the truncated APDS2-causal p85 alpha variant was very abundant in IRS1 and IRS2 precipitates, even in the absence of insulin stimulation. These important findings add in an interesting way possible mechanistic explanation for the growing number of APDS2 patients described with features of SHORT syndrome.

Strengths:

Based on state-of-the-art functional investigation the authors propose indicating a loss-of-function activity of the APDS2-disease causing p85 alpha variant in preadipocytes providing a possible mechanistic explanation for the growing number of APDS2 patients described with features of SHORT syndrome.

Weaknesses:

Related to Figure 1: PIK3R1 expression not only by Western blotting but also by quantifying

the RNA transcripts, e.g. mutant and wildtype transcripts, was not performed. RNA expression analysis would further strengthen the suggested impaired stabilization/binding.

Related to Figure 2: As mentioned by the authors in the manuscript the expression of p110 delta but also p110 beta in 3T3-L1 preadipocytes ectopically expressing p85 alpha variants has not been analyzed.

Furthermore, a direct comparison of the truncated APDS2-causal p85 alpha variant with SHORT syndrome -causal p85 alpha variants in regard to pAKT level, and p85 alpha expression level has not been performed.

These investigations would further strengthen the data.

Related to Figure 3:

The E489K and Y657X p85 alpha variants should be also tested in combination with p110 delta in the PI3K activity in vitro assay. This would help to further decipher the overall impact, especially of the E489K variant.

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